

Problem #3: Mnemonic Major System

Solution:

In the Mnemonic Major System, each digit corresponds to a **consonant sound** (which might be represented by one or more consonant letters). The vowels (*a, e, i, o, u*), as well as the semivowels (*y, w*) are not encoded in the numbers; rather they are used appropriately so as to form meaningful words from the consonant sequences derived from the number. Other rules are as follows:

- The letter *h* is also ignored during encoding.
- The pronunciation of the letter is considered during the encoding. Thus, the sequence *ph* in the word *elephant* is encoded as 8, which is for the sound *f*.

The mappings of the sounds to the digits are as follows.

Digit	Consonant Sound (in IPA)
0	/s/ /z/
1	/t/ /d/ /θ/ /ð/
2	/n/
3	/m/
4	/r/
5	/l/
6	/ʃ/ /ʒ/ /tʃ/ /dʒ/
7	/k/ /g/
8	/f/ /v/
9	/p/ /b/

Mnemonic Phrases	Numbers	Mnemonic c Phrases	Numbers
short striped zebra	64,101,491,094	cat dog	7,117
moon cash	3,276	grey elephant	745,821